

NovaCorp People Analytics Challenge

Accenture x SUBAA — August 2026

1. NovaCorp Overview

NovaCorp is a 12,000-person Australian financial services firm operating across seven departments: Retail Banking, Wealth Management, Insurance, Technology, Risk & Compliance, Corporate Operations, and Executive Leadership.

The firm has expanded through a series of acquisitions over the past three years, integrating several smaller entities into its existing workforce. As a result, its employee population spans multiple legacy systems, employment histories, and organisational cultures. NovaCorp does not currently have a dedicated people analytics function. The datasets provided to you represent the first consolidated export of HR data across the merged business.

This challenge explores themes that Accenture considers strategically relevant to organisations navigating workforce and talent strategy in financial services.

2. The Business Problem

NovaCorp's Chief Human Resources Officer (CHRO) has engaged your team to help understand and address an annual people cost estimated at **approximately \$42 million**.

Finance has decomposed this figure into three components:

Component	Estimated Range	Description
Regrettable attrition	~\$22–25M	Replacement cost of high-value voluntary departures
Disengagement-driven productivity loss	~\$12–15M	Productivity reduction for persistently disengaged employees

Component	Estimated Range	Description
Hiring inefficiency	~\$4–6M	Cost premium of agency hiring and poor-match early attrition

Your task is **not** to validate the \$42 million figure. It is to identify what is driving it, determine which components are most tractable to intervention, and advise the CHRO and CFO on where to prioritise.

The CHRO will be in the room at the 21 August presentation. They will ask hard questions. Your recommendation should be one they can act on.

3. The Challenge

This is a two-week datathon run in partnership between Accenture and SUBAA (Sydney University Business Analytics Association).

- Teams work independently — there are no assigned mentors during the challenge period
- **Finalist selection:** SUBAA will shortlist **4 finalist teams** from all submitted decks. Selection is based on the quality of the full presentation you submit by the deadline
- **Finals day:** 21 August 2026 at Accenture Sydney — the 4 finalist teams will receive a mentoring session at lunch, then present to a panel of Accenture judges in the afternoon

The absence of a mentor during the challenge is intentional. Your team's ability to navigate the problem independently is part of what is being assessed.

4. The Datasets

You are provided with four CSV files, all joinable via `employee_id`. Together, they cover a two-year observation window from **1 January 2024 to 31 December 2025**.

employees.csv — 13,403 rows

The full historical employee roster. This file includes **both active and departed employees** who were on the NovaCorp payroll at any point during the observation window.

Field	Type	Description
<code>employee_id</code>	string	Unique employee identifier
<code>name</code>	string	Employee full name
<code>hire_date</code>	date	Date employment commenced
<code>exit_date</code>	date	Date of departure; null if employee is still active
<code>status</code>	string	<code>active</code> or <code>departed</code>
<code>department</code>	string	Business division
<code>role_family</code>	string	Functional grouping: Operations-Processing, Technology, Client-Advisory, Risk-Compliance, Corporate-Support, Executive; Management applies at L5 and below
<code>role_level</code>	integer	Seniority level 1–8 (1 = Entry/Frontline IC; 8 = CEO)
<code>job_title</code>	string	Full job title corresponding to the employee's role family and level
<code>acting_appointment</code>	boolean	Whether the employee is in an acting or interim role; applies to L2–L4 employees
<code>salary</code>	float	Annual base salary in AUD; excludes superannuation, bonus, and equity

Field	Type	Description
compa_ratio	float	Salary as a proportion of the role-level band midpoint (1.0 = midpoint)
gender	string	Self-reported gender identity
age_band	string	Five-year age brackets
cultural_background	string	Self-reported cultural and ethnic background (ABS classification framework)
contract_type	string	Full-time, Part-time, Fixed-term, or Casual
hipo_flag	boolean	Whether the employee has been identified as high-potential through the formal talent review process
promotion_eligible	boolean	Whether the employee is currently eligible for promotion based on tenure and performance criteria
manager_id	string	The employee_id of the employee's direct manager; null for the most senior level
hire_source	string	How the employee was recruited: agency, direct, referral, graduate, or acquisition
legacy_entity_code	string	Acquisition origin: Entity_A, Entity_B, Entity_C, or NovaCorp-Origin
data_source_system	string	The source HRIS system from which this record was exported

Field	Type	Description
<code>days_to_fill</code>	integer	Days from vacancy opening to hire
<code>tenure_months</code>	integer	Months from hire date to the end of the observation window or to exit date, whichever applies; calculated using calendar months (not days ÷ 30), inclusive of the start month

Note: `employees.csv` includes both active and departed employees. Before conducting analysis, consider which population is appropriate for your question and document your reasoning. The choice matters.

attrition_log.csv — approximately 1,400 rows

Records of all employee departures that occurred during the observation window.

Field	Type	Description
<code>employee_id</code>	string	Links to <code>employees.csv</code>
<code>exit_date</code>	date	Date of last employment
<code>exit_type</code>	string	<code>voluntary</code> or <code>involuntary</code>
<code>stated_exit_reason</code>	string	Reason recorded by HR at the time of the exit interview. Note: research consistently shows 40–60% of exit-interview reasons do not reflect the primary driver when triangulated against longitudinal engagement data. Treat this field as one signal, not ground truth.

Field	Type	Description
<code>notice_period_served</code>	boolean	Whether the employee worked through their notice period
<code>regrettable_flag</code>	boolean	HR's classification of whether the departure is considered regrettable to the organisation
<code>performance_band_at_exit</code>	string	The employee's performance rating at their most recent review prior to departure
<code>salary_at_exit</code>	float	Annual base salary at the time of departure; equals the <code>salary</code> field in <code>employees.csv</code> for the same employee — no intra-window salary adjustment events are recorded
<code>manager_id_at_exit</code>	string	The <code>employee_id</code> of the employee's manager at the time of departure
<code>pathway</code>	string	<code>push</code> (involuntary or managed exit) or <code>pull</code> (employee-initiated, opportunity-driven); derived from exit interview and HR classification. Push exits were managed or required by the organisation; pull exits were self-initiated by the employee in response to an outside opportunity.

Note: `regrettable_flag`, `performance_band_at_exit`, and `stated_exit_reason` are retrospective judgements recorded by HR after the departure event. Consider carefully how you use these fields in your analysis.

engagement.csv — approximately 56,000 rows

Five waves of employee engagement survey data collected across the observation window.

Note: Employees hired after Wave 5 or who departed before Wave 1 may have fewer than five records; a small number have none.

Field	Type	Description
employee_id	string	Links to employees.csv
wave_number	integer	Survey wave (1 through 5)
survey_date	date	Date the survey was completed or attempted
response_flag	boolean	True if the employee responded; False if the survey was issued but not completed
manager_effectiveness	float	1–5 scale; null where response_flag is False
psychological_safety	float	1–5 scale; null where response_flag is False
recognition	float	1–5 scale; null where response_flag is False
career_development	float	1–5 scale; null where response_flag is False
senior_leadership_trust	float	1–5 scale; null where response_flag is False
purpose_meaning	float	1–5 scale; null where response_flag is False
wellbeing	float	1–5 scale; null where response_flag is False

Field	Type	Description
<code>confidence_in_role_future</code>	float	1–5 scale; null where <code>response_flag</code> is <code>False</code>

Note: Rows where `response_flag` is `False` are intentional inclusions — they represent employees who were issued the survey but did not respond. Do not discard these rows without first considering what their presence might tell you.

performance.csv — approximately 35,000 rows

Performance review records covering the observation window.

Field	Type	Description
<code>employee_id</code>	string	Links to <code>employees.csv</code>
<code>review_date</code>	date	Date the performance review was conducted
<code>performance_rating</code>	string	Rating assigned: <code>Outstanding</code> , <code>High Performer</code> , <code>Meets Expectations</code> , <code>Below Expectations</code> , or <code>Unsatisfactory</code>
<code>review_cycle</code>	string	Identifier for the review period
<code>promotion_recommendation</code>	boolean	Whether the reviewing manager recommended the employee for promotion
<code>goal_achievement_score</code>	float	Numeric goal attainment score on a 0–100 scale
<code>reviewer_id</code>	string	The <code>employee_id</code> of the manager who conducted the review

Note: The 2025-H2 performance review cycle (typically conducted November–December) is not included in this dataset. The most recent review for most employees is from the 2025-H1 or 2024-H2 cycle.

5. Data Quality

These four files are direct exports from NovaCorp's HR information systems, which span multiple legacy platforms following several corporate acquisitions over the past three years.

As with all production HR data, these exports contain characteristics typical of enterprise HRIS environments — including fields with incomplete population, historical records carried over from system migrations, and formatting variations that reflect different data entry practices across time and source systems.

The datasets are generated with fixed parameters. Teams should not attempt to reverse-engineer the generation process to obtain information not discoverable through standard analytical methods.

We recommend treating data preparation as a substantive analytical step, not a preprocessing formality. The decisions you make about how to handle data quality — which records to include or exclude, how to handle missing values, how to address inconsistencies — will be visible in your analysis. Documenting those decisions is part of demonstrating analytical rigour.

6. Analysis Guidance

The \$42M is a starting point, not a conclusion. The benchmark constants Finance used to derive that figure are listed below. Your job is to identify what is driving those costs and where NovaCorp should focus first.

Assumption	Value
Replacement cost multiplier	1.5× annual base salary
Backfill rate	85% of vacated positions filled
Disengagement productivity loss	15% of base salary per year
Superannuation on-cost	12.0% of base salary (1 July 2025 legislated rate)

Assumption	Value
Agency fee rate	18% of first-year base salary
Direct hire benchmark	\$5,500 per hire (fully loaded)

Cross-referencing the datasets is expected. Findings based on a single file will typically be less robust than findings that triangulate across multiple sources. The four datasets are designed to be used together.

Different teams will reach different conclusions. This is expected, and it will not be penalised. There is no single correct answer to what is driving NovaCorp's \$42 million people cost. Judges assess the quality of your reasoning, the rigour of your evidence, and the credibility of your recommendation — not whether your conclusion matches any predetermined outcome.

Limitations are part of the analysis. Acknowledging what your analysis cannot conclude — due to data availability, methodological constraints, or the nature of the evidence — is a mark of analytical maturity. Teams that present findings without limitations will be scrutinised more closely in Q&A, not less.

Think about your audience. The CHRO is your primary audience at the 21 August final. Every slide, every figure, and every recommendation should be calibrated to what they can act on, not what is technically interesting to produce.

7. Submission Requirements

Primary deliverable: A slide deck of up to 15 slides in PDF format (no limit on appendix slides)

Your audience is an **HR leader**, not a data scientist: make insights easy to understand and clearly actionable.

Additional deliverables: Optional but welcome. If your team produces a dashboard, Power BI report, website, or other interactive output, attach it with your submission.

Quantify your findings wherever possible — \$, % or headcount. Judges will ask: what happens if NovaCorp acts on your recommendation?

Submission deadline: Upload your submission to the shared Google Drive by **15 August 2026, 11:59pm**.

8. Evaluation Criteria

Your submission will be assessed across six dimensions:

Dimension	Weight
Ethics & Responsible Practice	20%
Analytical Rigour & Evidence Quality	25%
Business Problem Framing	15%
Financial Impact Quantification	15%
Recommendation Actionability	15%
Communication & Narrative Coherence	10%

9. Timeline

Date	Milestone
5 August, 11am–12pm	Virtual induction — Accenture presents the case and dataset briefing
4 August	Datasets released to all teams via the shared Google Drive
15 August, 11:59pm	Submission deadline — upload deck (and any additional deliverables) to the shared Google Drive
15–20 August	SUBAA reviews submissions and notifies the 4 finalist teams
21 August, 12pm–1pm	Finalist lunch + mentor check-in at Accenture Sydney (15–30 min per team)

Date	Milestone
21 August, 1pm–2:45pm	Finalist working session — teams implement feedback
21 August, 3pm–5pm	Final presentations at Accenture Sydney + winner announcement

There is no mid-challenge check-in or mentor touchpoint before the submission deadline. Manage your team's timeline independently.

10. Contact and Support

Direct general questions about the challenge format or dataset structure to SUBAA via the shared Google Drive or the Accenture contact provided at induction. Questions about analytical direction should be saved for your team.

NovaCorp is a fictional organisation created for this challenge. All data is synthetic. Any resemblance to real individuals, companies, or events is coincidental. Document version: August 2026. Accenture × SUBAA NovaCorp People Analytics Challenge.